

PARAINEN UTO, Finland

WMO: 029810

Lat: 59.784N

Lon: 21.368E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.3	-9.7	-14.9	1.0	-11.7	-12.5	1.3	-8.5	19.1	1.2	17.3	1.1	7.4	70	0.814

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	23.2	19.7	21.9	18.9	20.8	18.1	20.4	22.6	19.4	21.4	18.4	20.3	5.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.3	22.0	18.5	13.3	21.0	17.4	12.5	19.9	58.7	22.5	55.2	21.6	52.0	20.4	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	15.5	14.0	-12.7	24.3	4.2	2.0	-15.7	25.8	-18.1	27.0	-20.5	28.1	-23.5	29.6		
(5)			-13.2	20.7	4.0	1.6	-16.1	21.9	-18.4	22.8	-20.7	23.8	-23.6	24.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-0.6	-1.7	-0.2	3.4	8.0	13.1	17.5	17.6	13.6	8.5	4.5	1.7	(6)
(7)		DBStd	7.30	3.68	4.27	3.03	1.95	2.63	2.36	2.52	2.22	2.35	2.88	2.97	3.41	(7)
(8)		HDD10.0	1729	329	327	316	197	73	3	0	0	2	63	164	256	(8)
(9)		HDD18.3	4108	587	560	574	447	319	158	46	40	142	306	414	514	(9)
(10)		CDD10.0	702	0	0	0	0	12	94	234	236	111	16	0	0	(10)
(11)		CDD18.3	40	0	0	0	0	0	0	22	18	1	0	0	0	(11)
(12)		CDH23.3	37	0	0	0	0	0	0	30	7	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.3	8.6	7.8	7.3	6.3	5.8	5.9	5.8	6.3	7.3	8.1	8.8	9.2	(14)
(15)	Precipitation	PrecAvg	568	47	36	33	27	28	44	50	61	60	64	64	55	(15)
(16)		PrecMax	699	79	77	70	54	63	82	128	140	178	131	148	98	(16)
(17)		PrecMin	443	8	3	6	6	5	9	3	9	13	5	7	14	(17)
(18)		PrecStd	66	19	18	15	13	18	18	35	30	39	29	30	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	4.7	5.6	10.1	17.8	20.8	25.9	19.7	14.7	10.2	7.7	(19)	
(20)			MCWB	5.3	3.6	3.9	6.9	13.6	16.1	21.4	20.8	16.8	13.6	9.3	7.2	(20)
(21)		2%	DB	5.1	3.9	4.6	8.4	15.2	18.7	24.0	22.7	18.5	13.7	9.7	7.0	(21)
(22)			MCWB	4.5	3.2	3.3	6.1	11.9	15.3	20.3	19.3	16.2	12.6	9.0	6.4	(22)
(23)		5%	DB	4.5	3.4	3.8	7.3	13.0	17.5	22.6	21.7	17.6	12.8	9.1	6.3	(23)
(24)			MCWB	3.8	2.7	2.7	5.4	10.4	14.6	19.5	18.6	15.5	11.7	8.4	5.5	(24)
(25)		10%	DB	3.7	2.8	3.1	6.2	11.6	16.5	21.3	20.8	16.7	12.0	8.2	5.6	(25)
(26)			MCWB	3.0	2.2	2.2	4.6	9.5	14.0	18.7	18.0	14.9	10.9	7.4	4.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	4.2	4.3	7.4	14.1	17.1	22.2	21.7	17.8	14.1	9.9	7.5	(27)
(28)			MCD B	5.8	4.3	5.0	9.1	17.2	19.4	24.6	23.4	19.1	14.3	10.0	7.6	(28)
(29)		2%	WB	4.7	3.4	3.6	6.4	12.2	16.0	20.9	20.2	16.7	12.7	9.2	6.5	(29)
(30)			MCD B	5.0	3.7	4.2	8.0	14.6	17.9	23.2	22.2	17.9	13.4	9.6	6.7	(30)
(31)		5%	WB	3.9	2.8	3.0	5.6	10.9	15.2	19.9	19.2	15.9	12.0	8.5	5.7	(31)
(32)			MCD B	4.4	3.2	3.6	7.0	12.7	17.0	22.2	21.0	17.1	12.7	9.0	6.1	(32)
(33)		10%	WB	3.1	2.3	2.4	4.8	9.7	14.4	18.9	18.4	15.2	11.2	7.6	5.0	(33)
(34)			MCD B	3.7	2.8	3.0	6.0	11.3	16.1	21.2	20.2	16.4	11.9	8.2	5.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	2.7	2.8	3.0	3.6	4.1	3.8	3.7	3.3	2.8	2.4	2.3	2.5	(35)
(36)			MCDBR	2.5	2.3	3.1	4.9	6.2	5.3	4.8	3.9	3.5	2.6	2.3	2.4	(36)
(37)			MCWBR	2.6	2.2	2.5	3.2	3.7	3.1	2.7	2.4	2.6	2.3	2.6	2.5	(37)
(38)		5% WB	MCD BR	2.5	2.2	2.9	4.8	5.9	4.9	4.4	3.5	3.2	2.5	2.3	2.2	(38)
(39)			MCWBR	2.7	2.3	2.5	3.3	3.8	3.1	2.7	2.5	2.6	2.5	2.6	2.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.284	0.317	0.334	0.373	0.368	0.362	0.390	0.382	0.352	0.337	0.312	N/A	(40)	
(41)		taud	2.192	2.261	2.318	2.327	2.426	2.485	2.430	2.439	2.504	2.404	2.219	N/A	(41)	
(42)		Ebn at Noon	492	681	796	826	860	872	833	812	780	658	445	N/A	(42)	
(43)		Edh at Noon	43	69	89	105	102	98	102	94	75	61	42	N/A	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.25	0.81	2.38	4.13	5.68	6.23	5.82	4.55	2.81	1.19	0.33	0.16	(44)	
(45)		RadStd	0.03	0.14	0.27	0.42	0.50	0.39	0.48	0.38	0.32	0.17	0.07	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	-116	N/A	N/A	N/A

Nomenclature: See separate page